# Path-Coloring Algorithms for Plane Graphs

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**Abstract**

A path coloring of a graph $G$ is a vertex coloring of $G$ such that each color class induces a disjoint union of paths. We present two efficient algorithms to construct a path coloring of a plane graph.

The first algorithm, based on a proof of Poh, is given a plane graph, and produces a path coloring of the given graph using three colors.

The second algorithm, based on similar proofs by Hartman and Škrekovski, performs a list-coloring generalization of the above. The algorithm is given a plane graph and an assignment of lists of three colors to each vertex, and it it produces a path coloring of the given graph in which each vertex receives a color from its list.

Implementations are available for all algorithms that are described.

## 1 Introduction

All graphs will be finite and simple. See West [12] for graph theoretic terms.

A graph is *planar* if it may be drawn in the plane by mapping each vertex to a point, and each edge to a simple arc between its incident vertices such that the arcs intersect only at their endpoints. A *face* is a region of the plane not containing any point of the

drawing; we'll refer to each face by the subgraph drawn on its border. A *(combinatorial) embedding* is an equivalence class of drawings determined by the circular ordering of edges around each vertex. A *plane graph* is a planar graph with a fixed embedding. Subgraphs of a plane graph are plane graphs with the inherited embedding.

A *path coloring* is a (vertex) coloring where each color class induces a disjoint union of paths. A graph $G$ is *path $k$-colorable* if $G$ admits a path $k$-coloring (a path coloring using at most $k$ colors). A *list assignment* $L$ of a graph $G$ maps each $v \in V(G)$ to a finite set of colors $L(v)$. An *$L$-coloring* (or *$L$-list-coloring*) is a coloring $c$ such that $c(v) \in L(v)$ for each $v \in V(G)$. A graph is *path $k$-choosable* if it admits a path $L$-coloring for every list assignment $L$ with at least $k$ colors in each list, i.e. $|L(v)| \geq k$ for each $v \in V(G)$.

Broere & Mynhardt conjectured [2, Conj. 16] that every planar graph is path 3-colorable. This was proven independently by Poh [9, Thm. 2] and Goddard [6, Thm. 1].

**Theorem 1.1** (Poh 1990, Goddard 1991)**.** *If $G$ is a planar graph, then $G$ is path 3-colorable.* □

It is easily shown that the "3" in Theorem 1.1 is best possible. In particular, Chartrand & Kronk [5, Section 3] gave an example of a planar graph whose vertex set cannot be partitioned into two subsets, each inducing a forest.

Hartman [7, Thm. 4.1] proved a list-coloring generalization of Theorem 1.1 (see also Chappell & Hartman [4, Thm. 2.1]). Essentially the same coloring technique was used by Škrekovski [11, Thm. 2.2b] to prove a result slightly weaker than Theorem 1.2.

**Theorem 1.2** (Hartman 1997)**.** *If $G$ is a planar graph, then $G$ is path 3-choosable.* □

We discuss two path coloring algorithms based on proofs of the above theorems.

In Section 2 we outline our graph representations and the basis for our computations of time complexity.

Section 3 covers an algorithm based on Poh's proof of Theorem 1.1. The algorithm is given a plane graph; it produces a path coloring of the given graph using three colors.

Section 4 covers an algorithm based on Hartman's proof of Theorem 1.2, along with the proof of Škrekovski mentioned above. The algorithm is given a plane graph and an assignment of a list of three colors to each vertex; it produces a path coloring of the given graph in which each vertex receives a color from its list.

Section 5 provides benchmark results for implementations of each algorithm [3]. The section also discusses how each algorithm may be modified to benefit from parallelism.

# 2 Graph Representations and Time Complexity

A plane graph $G$ is *triangulated* (or *maximal*) if it is connected and $n(G) \leq 2$, or if every face of $G$ is a *triangle* (a length 3 cycle). We say $G$ is *weakly triangulated* if all of its interior (bounded) faces are triangles (a triangulated graph is also weakly triangulated).

The input for each algorithm will be a weakly triangulated plane graph $G$ with $n$ vertices and $m$ edges. The input size will be $n$. By Euler's formula $m \leq 3n - 6$, so it is

equivalent use input size $m$. Planar graphs may be plane embedded and triangulated (by adding edges) in $\mathcal{O}(n)$ time, see e.g. Hopcroft and Tarjan [8] or Boyer and Myrvold [1].

We will represent $G$ for computation via *adjacency lists*: a list, for each vertex $v$, of the neighbors of $v$. A vertex will be represented by an integer $0 \ldots n-1$. The order of each list represents the embedding. In algorithms $N(v)$ will refer to the adjacency list.

Let $v \in G$ and $u \in N(v)$. Define $\text{next}_v(u) \in N(v)$ to be the neighbor counter-clockwise from $u$ around $v$, and $\text{prev}_v(u) \in N(V)$ to be the neighbor clockwise from $u$. In algorithms ${}^v u$ will be a reference to $u$'s entry in $N(v)$, i.e. ${}^v u$ represents both $u$ and the position of $u$ within $v$'s adjacency list; define $\text{next}({}^v u) = {}^v(\text{next}_v(u))$ and $\text{prev}({}^v u) = {}^v(\text{prev}_v(u))$.

Adjacency lists naturally allow $\text{next}({}^v u)$ to be computed in constant time. However, the linear time complexity of the algorithm discussed in Section 4 depends on the ability to compute $\text{prev}({}^v u)$ and the *back reference* ${}^u v$ from ${}^v u$ in constant time as well. Representing $N(v)$ via an array or doubly linked list solves computing both $\text{next}({}^v u)$ and $\text{prev}({}^v u)$, but back references require more information. We define an *augmented adjacency list* to be an adjacency list that stores the pair $(u, {}^u v)$ for each neighbor $u$ of $v$. Given an adjacency list representation, an augmented representation may be constructed in $\mathcal{O}(m)$ time.

**Algorithm 2.1:** Augment Adjacency Lists

**Input.** An adjacency list representation $N$ of a graph $G$.

**Output.** An augmented adjacency list representation $N^*$ of $G$ such that for each $v \in G$, the neighbors of $v$ appear in $N^*(v)$ in the same order they appear in $N(v)$.

**Step 1.** For each $v \in G$, construct an augmented adjacency list $N^*(v)$ with the same ordering as $N(v)$, leaving the ${}^u v$ portion of each entry $(u, {}^u v)$ in $N^*(v)$ uninitialized.

**Step 2.** For each vertex $v$, construct a list $\text{Wrk}(v)$ of vertex-reference pairs with capacity $\deg(v)$. For each $v$ from 0 to $n-1$, iterate through $N^*(v)$. For each neighbor $u$ in $N^*(v)$, append the pair $(v, {}^v u)$ to $\text{Wrk}(u)$, where ${}^v u$ is a reference to $u$'s entry $N^*(v)$.

**Step 3.** For each $v$ from $n-1$ to 0, iterate through $\text{Wrk}(v)$. Upon reaching a pair $(u, {}^u v)$ in $\text{Wrk}(v)$, the last element of $\text{Wrk}(u)$ will be $(v, {}^v u)$; see below. Lookup the entry ${}^v u$ in $N^*(v)$ and initialize the back reference ${}^u v$, and vice versa. Pop $(v, {}^v u)$ from $\text{Wrk}(u)$.

After completing Step 2 in Algorithm 2.1 each list $\text{Wrk}(v)$ will contain $(u, {}^u v)$ for each neighbor $u$ of $v$, sorted in increasing order by $u$. Suppose that $v$ is the current vertex at a given iteration of the loop in Step 3. For each edge $uw \in E(G)$ such that $u < w$ and $v < w$, prior iterations of the loop will have initialized the back references for $(w, {}^w u)$ in $N^*(u)$ and $(u, {}^u w)$ in $N^*(w)$, and removed $(w, {}^w u)$ from $\text{Wrk}(u)$. Therefore for each $(u, {}^u v)$ in $\text{Wrk}(v)$, the list $\text{Wrk}(u)$ will only contain entries $(w, {}^w u)$ where $w \leq v$.

# 3 Path Coloring

In this section we describe a path 3-coloring algorithm based on a proof by Poh [9, Thm. 1].

Let $C$ be a cycle in a connected plane graph $G$. If $u, v \in C$, then we define $C[u, v]$ to be the clockwise $u, v$-path along $C$; if $u = v$, then $C[u, u] = u$. Define $\overline{\text{int}}(C)$ to be the subgraph of $G$ consisting of $C$ and all vertices and edges interior to $C$; equivalently, $\overline{\text{int}}(C)$ is the maximal subgraph of $G$ with outer face $C$. Define $\text{int}(C) = \overline{\text{int}}(C) - C$.

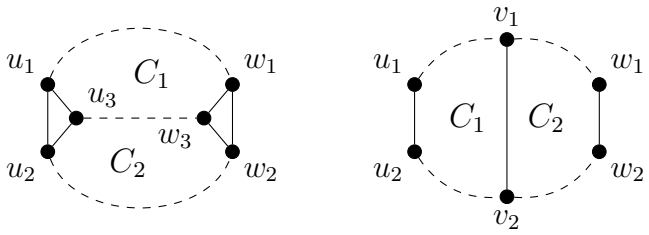



Figure 1: The proof of Lemma 3.1 when $C$ is induced (left), or has a chord (right).

Let $P$ be a path in a plane graph $G$. If $u, v \in P$, then we define $P[u, v]$ to be the $u, v$-path in $P$. We also define $\overline{\text{int}}(P) = P$ and $\text{int}(P)$ to be empty.

**Lemma 3.1** (Poh 1990)**.** *Let $G$ be a biconnected, weakly triangulated plane graph with outer face $C$. Let $c : V(C) \to S \subsetneq \{1, 2, 3\}$ be a 2-coloring of $C$ such that each color class induces a nonempty path. There exists an extension of $c$ to a path 3-coloring $c : V(G) \to \{1, 2, 3\}$ such that for each $v \in G - C$, if $vu \in E(G)$ with $u \in C$, then $c(v) \neq c(u)$.*

*Proof.* If $n(G) \leq 3$, then $G = C$. We proceed by induction on the order of $G$.

Suppose that $G \neq C$. Let $P_1, P_2$ be the two paths induced by the 2-coloring of the outer face $C$. We may write $P_1 = C[u_1, w_1]$ and $P_2 = C[w_2, u_2]$ for vertices $u_1, u_2, w_1, w_2 \in C$.

Suppose that $C$ is an induced subgraph of $G$. Take $u_3, w_3 \in G - C$ such that the cycles $u_3, u_2, u_1$ and $w_3, w_1, w_2$ are faces of $G$; note that $u_3$ and $w_3$ are uniquely determined, but it may be that $u_3 = w_3$. Since $C$ is induced, $G - C$ is connected. Let $P_3$ be an induced $u_3, w_3$-path in $G - C$. Color each vertex of $P_3$ with $d \in \{1, 2, 3\} - S$. Let $C_1 = P_1 \cup P_3 \cup u_3u_1 \cup w_3w_1$ and $C_2 = P_2 \cup P_3 \cup u_3u_2 \cup w_3w_2$, see Figure 1 (left). By the inductive hypothesis, the 2-coloring of $C_1$ extends to a path 3-coloring of $\overline{\text{int}}(C_1)$. Similarly, the coloring of $C_2$ extends to a path 3-coloring of $\overline{\text{int}}(C_2)$. Together these colorings form a 3-coloring of $G$. Since $C$, $P_3$, $\text{int}(C_1)$, and $\text{int}(C_2)$ partition $V(G)$, and no vertex shares a color with a neighbor in a different partition, this 3-coloring is a path coloring satisfying the lemma.

Suppose that $C$ is not an induced subgraph. Then there exists an edge $v_1v_2 \in E(G) - E(C)$ with $v_1 \in P_1$ and $v_2 \in P_2$. Let $C_1 = C[v_2, v_1] \cup v_1v_2$ and $C_2 = C[v_1, v_2] \cup v_1v_2$, see Figure 1. By the inductive hypothesis, $c$ extends to path 3-colorings of $\overline{\text{int}}(C_1)$ and $\overline{\text{int}}(C_2)$. Together these colorings form a path 3-coloring of $G$. □

Let $G$ be a triangulated plane graph. We may trivially path 2-color the outer triangle. By Lemma 3.1 this extends to a path 3-coloring of $G$. If $G$ is an arbitrary planar graph, we may compute an embedding and add edges to produce a triangulated plane graph $G'$. Theorem 1.1 follows immediately as any path coloring of $G'$ is also a path coloring of $G$.

Let $G$, $C$, $P_1$, and $P_2$ be as in Lemma 3.1, and assume that $C$ is induced in $G$. For a subgraph $H$ of $G$, let $N(H)$ be the set of vertices in $G - H$ that have a neighbor in $H$. Let $G_1$ be the subgraph of $\overline{\text{int}}(C)$ induced by $V(P_1) \cup N(P_1) - V(P_2)$, see Figure 2. Let $C_1$ be the outer face of $G_1$. Algorithm 3.2 implements Poh's algorithm in linear time using $P_3 = C_1 - P_1$ as the $u_3, w_3$-path. Observe that $P_3$ is induced in $G$: $G_1$ is induced,

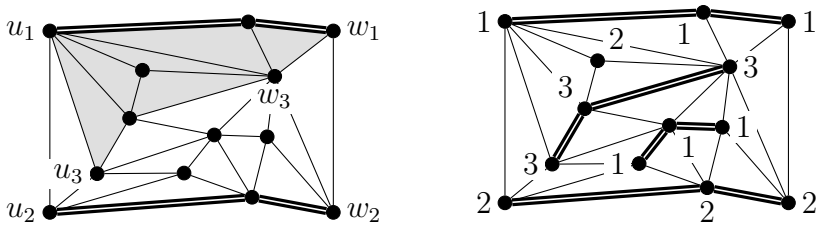



Figure 2: An example setup for Poh's algorithm highlighting the subgraph $G_1$ induced by $V(P_1) \cup N(P_1) - V(P_2)$ (left), and the path 3-coloring output from Algorithm 3.2 (right).

so there is no chord of $P_3$ exterior to $C_1$, and each $v \in P_3$ has a neighbor in $P_1$, so there is no chord of $P_3$ interior to $C_1$. To find $P_3$, we walk counter-clockwise from $u_3$ along $C_1$.

Define $p_1 : N(P_1) \to V(P_1)$ to be the map sending each $v \in N(P_1)$ to its neighbor in $P_1$ that is farthest from $u_1$, i.e. $V(C[p_1(v), w_1]) \cap N(v) = \{p_1(v)\}$.

**Algorithm 3.2:** Path 3-Color

**Input.** Let $C$ be an induced cycle in a weakly triangulated plane graph $G$ represented via adjacency lists. Let $c$ be a length $n$ array representing a 2-coloring of $C$ such that each color class induces a nonempty path, respectively $P_1 = C[u_1, w_1]$ and $P_2 = C[w_2, u_2]$, where $u_1, u_2, w_1, w_2 \in C$. Assume that $c(v) = 0$ for each $v \in \text{int}(C)$. Let $S$ be a length $n$ array, and $\ell_1$ an integer, such that for $v \in \text{int}(C)$, $S(v) = \ell_1$ if and only if $v \in N(P_1)$.

The concrete input will be $G$, $c$, and $S$, along with the tuple $(u_1, {}^{u_1}u_2, \ell_1)$.

**Output.** For each vertex $v \in \text{int}(C)$, a color in $\{1, 2, 3\}$ will be assigned to $c(v)$ such that $c$ represents a path 3-coloring of $\overline{\text{int}}(C)$ extending the original 2-coloring of $C$. Moreover, if $v \in \text{int}(C)$ has a neighbor $u \in C$, then $c(v) \neq c(u)$.

**Procedure.** Let ${}^{u_1}u_3 := \text{next}({}^{u_1}u_2)$. It may be useful to refer to Figures 2 and 3.

**Base case.** If $c(u_3) \neq 0$, then $u_3 = w_2$ or $u_3 = w_1$. Thus $C$ is a face and we are done.

**Recursive case.** Suppose that $c(u_3) = 0$, i.e. $u_3 \in \text{int}(C)$. Let $d \in \{1, 2, 3\} - \{c(u_1), c(u_2)\}$. Let $\ell_3 := \max_{v \in G} S(v)+1$. We'll walk the induced $u_3, w_3$-path $P_3 = C_1 - P_1$, where $C_1$ is the outer face of the subgraph $G_1$ of $\overline{\text{int}}(C)$ induced by $V(P_1) \cup N(P_1) - V(P_2)$.

**Step 1.** Let $a := u_1$ and $z := u_3$; here $z$ stores the current vertex of $P_3$, and $a$ stores the neighbor of $z$ clockwise from $z$ around $C_1$. Let $t :=$ NULL and ${}^t x :=$ NULL; here $t$ and ${}^t x$ store the last edge $tx$ visited where $t \in P_3$ and $x \in P_2$ (the first such edge is $u_3u_2$).

**Loop 2.** We'll walk counter-clockwise along $C_1$, coloring one vertex $z$ per iteration, and halting after $z = w_3$. Color $c(z) \leftarrow d$. Find ${}^z a$ in $N(z)$ and let ${}^z v := \text{next}({}^z a)$. Let $b :=$ NULL; here $b$ will store neighbor of $z$ in $P_3$ that is counter-clockwise from $z$ around $C_1$, if it exists. Let ${}^z y :=$ NULL; here ${}^z y$ will store the last edge $zy$ visited where $y \in P_1$.

**Loop 2.1.** We'll first consider the neighbors of $z$ from $\text{next}_z(a)$ to $p_1(z)$ counter-clockwise. Each iteration will handle one neighbor ${}^z v$. We'll make a recursive call for each induced cycle $C'$ we discover with $V(C') \subseteq V(P_2) \cup V(P_3)$.

**Case 2.1.1.** Suppose that $c(v) = 0$, i.e. $v \in \text{int}(C)$. If $S(v) = \ell_1$ and $b =$ NULL, then $v \in P_3$ and we set $b \leftarrow v$. Set $S(v) \leftarrow \ell_3$ and ${}^z v \leftarrow \text{next}({}^z v)$. Repeat Loop 2.1.

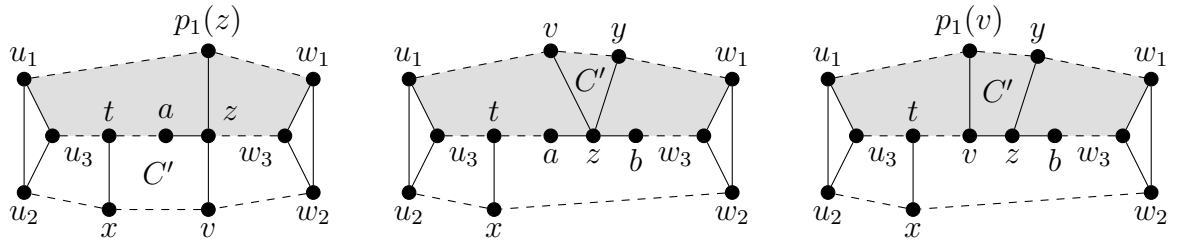



Figure 3: A sketch of Case 2.1.2 (left), Case 2.2.2 (middle), and Case 2.2.3 (right) in Algorithm 3.2. In each sketch, the highlighted region represents $G_1$.

**Case 2.1.2.** Suppose that $c(v) = c(u_2)$, i.e. $v \in P_2$. If $t \neq$ NULL, make a recursive call with input $(t, {}^tx, \ell_3)$ to color $\overline{\text{int}}(C')$, where $C' = P_3[t, z] \cup C[v, x] \cup tx \cup zv$; refer to Figure 3 (left). Set $t \leftarrow z$, ${}^tx \leftarrow {}^zv$, and ${}^zv \leftarrow \text{next}({}^zv)$. Repeat Loop 2.1.

**Case 2.1.3.** Suppose that $c(v) = c(u_1)$, i.e. $v = p_1(z) \in P_1$. Break Loop 2.1. If $v = a$, skip to Step 2.3. Otherwise, set ${}^zy \leftarrow {}^zv$, ${}^zv \leftarrow \text{next}({}^zv)$, and proceed to Loop 2.2.

**Loop 2.2.** We'll now consider the neighbors of $z$ from $\text{next}_z(p_1(z))$ to $a$ counterclockwise. Each iteration will handle one neighbor ${}^zv$. We'll make a recursive call for each induced cycle $C'$ we discover with $V(C') \subseteq V(P_1) \cup V(P_3)$. Observe that ${}^zy \neq$ NULL.

**Case 2.2.1.** Suppose that $c(v) = 0$, i.e. $v \in \text{int}(C_1)$. Assign $S(v) \leftarrow \ell_3$ and ${}^zv \leftarrow \text{next}({}^zv)$. Repeat Loop 2.2.

**Case 2.2.2.** Suppose that $c(v) \neq 0$ and $v \neq a$, i.e. $v \in P_1$. Make a recursive call with input $(z, {}^zy, \ell_3)$ to color $\overline{\text{int}}(C')$, where $C' = C[v, y] \cup z \cup zy \cup zv$; see Figure 3 (middle). Set ${}^zy \leftarrow {}^zv$ and ${}^zv \leftarrow \text{next}({}^zv)$, then repeat Loop 2.2.

**Case 2.2.3.** Suppose that $v = a$. Make a recursive call with input $(z, {}^zy, \ell_3)$ to color $\overline{\text{int}}(C')$, where $C' = C[p_1(v), y] \cup P_3[v, z] \cup vp_1(v) \cup zy$; see Figure 3 (right). Break Loop 2.2 and proceed to Step 2.3.

**Step 2.3.** If $b \neq$ NULL, then assign $a \leftarrow z$, $z \leftarrow b$, and repeat Loop 2. Otherwise, $z = w_3$ and $c$ is a path 3-coloring of $\overline{\text{int}}(C)$, so we are done.

Given a triangulated plane graph $G$ represented via adjacency lists, we can apply Algorithm 3.2 as follows. First, create length $n$ integer arrays for $c$ and $S$, initializing each entry to zero. Let $C = v_1, v_2, v_3$ be the outer triangle of $G$, labeled in clockwise order. Color $c(v_1) \leftarrow 1$, $c(v_2) \leftarrow 2$, and $c(v_3) \leftarrow 2$. Iterate through $N(v_1)$ and mark $S(v) \leftarrow 1$ for each neighbor $v$ of $v_1$. Then $G$, $c$, $S$, $(v_1, {}^{v_1}v_3, 1)$ form a valid input for Algorithm 3.2.

During Algorithm 3.2 we iterate through $N(v)$ for each $v \in G$ exactly twice: once to locate ${}^za$ when $z = v$ is colored at the start of Loop 2, and once to iterate through $N(z)$ in Loops 2.1 and 2.2. Therefore the time complexity is $\mathcal{O}(n)$.

# 4 Path List-Coloring

In this section we describe an algorithm to path color plane graphs such that each vertex receives a color from a specified list. Hartman showed that this is always possible when

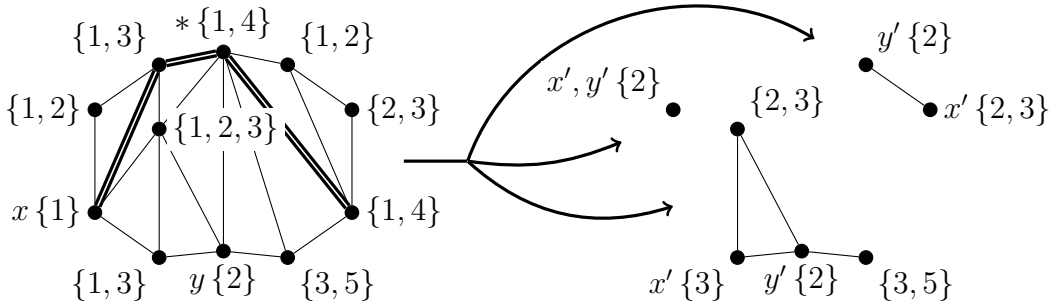



Figure 4: Coloring and removing a path $P$ as in Hartman's proof [7, Lemma 4.2].

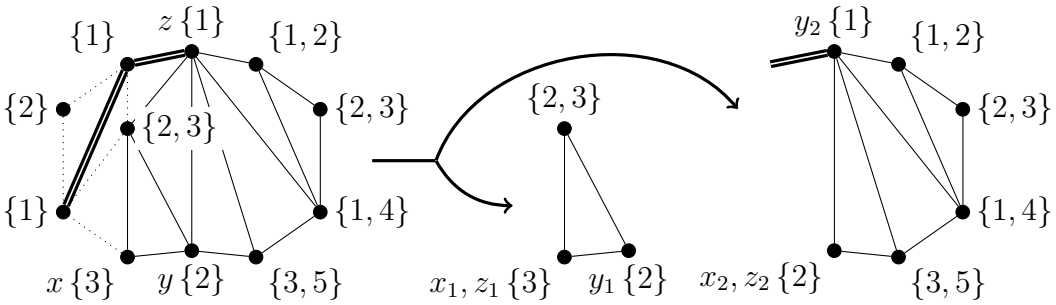



Figure 5: Coloring and removing a path $P$ as in the proof of Lemma 4.1 (Algorithm 4.2).

each vertex is given a list of 3 colors [7, Thm. 4.1]. Around the same time Škrekovski proved a weaker result using the same coloring strategy [11, Thm. 2.2b].

In brief, the Hartman-Škrekovski algorithm takes as its input a weakly triangulated plane graph $G$ with outer face $C$, a list assignment $L$, and two vertices $x, y \in C$. It assumes that $|L(x)| \geq 1$, $|L(y)| \geq 1$, $|L(v)| \geq 2$ for each $v \in C - \{x, y\}$, and $|L(v)| \geq 3$ for each $v \in G - C$. An induced path $P$ is colored with $d \in L(x)$, proceeding from $x$ towards $y$ along $C[x, y]$, such that for each $v \in P$, $d \in L(v)$. Starting from $P = x$, the path is colored as follows: let $z$ be end of $P$ opposite $x$, and define $T_z = \{v \in N(z) \cap V(C[z, y]) : d \in L(v)\}$; if $T_z \neq \emptyset$, then the $v \in T_z$ farthest clockwise along $C$ is colored and added to $P$, and this process is repeated with $z = v$. After coloring $P$, the algorithm is applied to each connected component of $G - P$, using the list assignment $L'$ defined by $L'(v) = L(v) - \{d\}$ if $v \in N(P)$, $L'(v) = L(v)$ if $v \notin N(P)$. See Figure 4 for an example.

In Lemma 4.1 below, we provide a different proof of Hartman's result, organized in such a way that it immediately describes a linear time algorithm for graphs represented as in Section 2. If $x = z$, then our $G, x, y$ match those in Hartman's [7, Lemma 4.2], but we assume that $G$ is biconnected; we follow a similar iterative procedure to color a path $P$ clockwise from $x$ along $C$, but we consider one edge at a time, and simultaneously remove $P$ as it is colored. If $x \neq z$, then $z$ is the farthest clockwise end of the path $P$

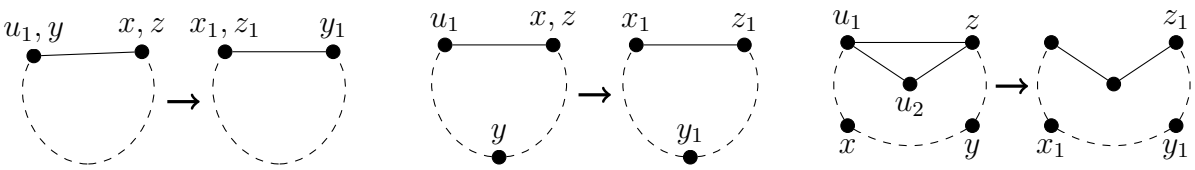



Figure 6: The proof of Lemma 4.1, Case 1 (left), Case 2 (middle), and Case 3.1 (right).

being colored; the rest of $P$ has already been removed, and our $x$ serves as Hartman's $x'$.

For those familiar with Hartman's proof: the one major difference between our algorithm and Hartman's occurs in Case 3.2.3.2 of the proof of Lemma 4.1 below. If we encounter a chord edge $zv \in E(G) - E(C)$ ($z$ is the current end of $P$) with $v \in C[z, y]$ and $L(z) \cap L(v) = \emptyset$ (i.e. $d \notin L(v)$, where $d$ is the color of $P$), then instead of continuing to search for neighbors of $z$ in $C[z, v]$ that we might add to $P$, we apply the inductive hypothesis such that $z$ now serves as "$y$" (see Figure 8 for a general illustration of this case). The edge from the vertex labeled $*$ to $y$ in Figure 4 is an example of such a chord: our algorithm halts $P$ at $*$, and divides the remaining subgraph at the chord, see Figure 5.

Given a coloring $c$ of a graph $G$, and $v \in G$, we define $\deg_c(v)$ to be $\deg(v)$ in the subgraph induced by $c(v)$'s color class, i.e. the number of neighbors that share $v$'s color.

**Lemma 4.1.** *Let $G$ be a biconnected, weakly triangulated plane graph with outer face $C$. Let $x, y, z \in C$ be vertices (not necessarily distinct) such that $z \in C[x, y]$. Let $L$ be a list assignment of $G$ such that*

$$|L(v)| = 1 \text{ for } v \in \{x, y, z\},$$
$$|L(v)| \geq 2 \text{ for } v \in C - \{x, y, z\},$$
$$|L(v)| \geq 3 \text{ for } v \in G - C,$$

*and if $v \in C[x, z] - z$, then $L(v) \cap L(z) = \emptyset$. There exists a path $L$-coloring $c$ of $G$ such that $\deg_c(x) \leq 1$, $\deg_c(y) \leq 1$, and $\deg_c(z) \leq 1$. Moreover, if $z = y$, or if $C[z, y] = z, y$ and $L(z) \cap L(y) = \emptyset$, then $\deg_c(z) = 0$.*

*Proof.* Suppose that $m = m(G) \leq 1$. Then $G = C$. We define a path $L$-coloring $c$ of $G$ as follows. Let $c(x) \in L(x)$ and $c(y) \in L(y)$ be the color in each respective list. If $m = 0$, then $n(G) = 1$ and $c$ trivially satisfies the lemma. Otherwise $m = 1$ and $n(G) = 2$. If $x \neq y$, then $c$ is a path $L$-coloring of $G$ with the constraints of the lemma. Otherwise the remaining $v \in G - y$ satisfies $|L(v)| \geq 2$, and we pick $c(v) \in L(v) - \{c(y)\}$.

We proceed by induction on $m$, the number of edges. Let $u_1, u_2, \ldots, u_k$ label the neighbors of $z$ in counter-clockwise order such that $u_1$ is the vertex immediately counter-clockwise from $z$ around $C$, and $u_k$ is the vertex clockwise from $z$ around $C$. In each of the subsequent cases we will apply the lemma separately to one or two subgraphs $G_1, G_2$ of $G$, with corresponding labeled outer face vertices $x_i, y_i, z_i \in G_i$ for $i = 1, 2$.

**Case 1.** Suppose that $u_1 = y$. Note that $z = x$ since $z \in C[x, y]$. Therefore $V(C[z, y]) = V(C)$ and $|V(C[z, y])| \geq 3$, so $C[z, y] \neq z, y$. Apply the lemma to find a path $L$-coloring $c$ of $G_1 = G$ with $x_1 = y$, $y_1 = x$, and $z_1 = y$, see Figure 6 (left).

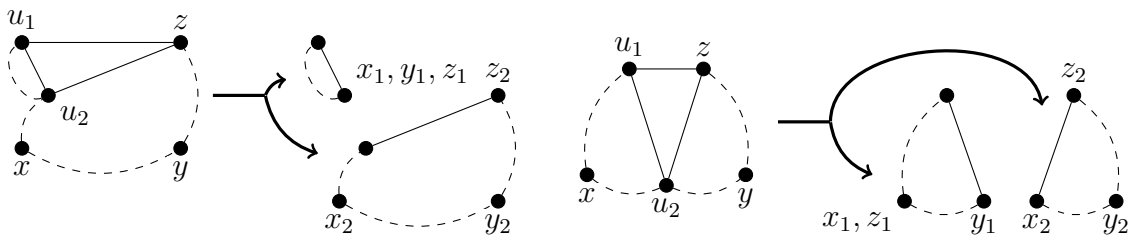



Figure 7: The proof of Lemma 4.1, Case 3.2.1 (left) and Case 3.2.2 (right).

**Case 2.** Suppose that $u_1 \neq y$ and $z = x$. Then $u_1 \in C - C[x, z]$ and $|L(u_1)| \geq 2$. Pick $d \in L(u_1) - L(z)$. Define $L'(u_1) = \{d\}$ and $L'(v) = L(v)$ for each $v \in G - u_1$. Apply the lemma to find a path $L'$-coloring of $G_1 = G$ with $x_1 = u_1$, $y_1 = y$, and $z_1 = z$, see Figure 6 (middle).

**Case 3.** Suppose that $u_1 \neq y$ and $z \neq x$. Our strategy will be to apply the inductive hypothesis to the *blocks* (maximal biconnected subgraphs) of $G - zu_1$. Because $u_1 \in C[x, z]$, we have $L(u_1) \cap L(z) = \emptyset$. Thus any path $L$-coloring of $G - zu_1$ is also a path $L$-coloring of $G$. There are three cases to consider, refer to Figures 6 (right), 7, and 8.

**Case 3.1.** Suppose that $u_2 \in G - C$. Define $L'(u_2) = L(u_2) - L(z)$ and $L'(v) = L(v)$ for each $v \in G - u_2$. Note that $|L'(u_2)| \geq |L(u_2)| - 1 \geq 2$. Let $C_1$ be the outer face of $G_1 = G - zu_1$, and observe that $C_1[x, z]$ is the union of $C[x, z] - u_1z$ with the path $u_1, u_2, z$. Since $|L'(u_2)| \geq 2$ and $L'(u_2) \cap L'(z) = \emptyset$, we may apply the inductive hypothesis to find a path $L'$-coloring of $G_1$ with $x_1 = x$, $y_1 = y$, $z_1 = z$.

**Case 3.2.** Suppose that $u_2 \in C$. Observe that $u_2$ is a *cut-vertex* of $G - zu_1$, i.e. $G - zu_1 - u_2$ is disconnected. Define $C_1 = C[u_2, u_1] \cup u_1u_2$ and $C_2 = C[z, u_2] \cup zu_2$. Let $G_1 = \overline{\text{int}}(C_1)$ and $G_2 = \overline{\text{int}}(C_2)$. The subgraphs $G_1$ and $G_2$ are the two blocks of $G - zu_1$. Note that if $k = 2$, then $G_1 = G - z$ and $G_2 = z, u_2$.

In each subsequent case we will apply the inductive hypothesis to find a path $L$-coloring $c_1$ of $G_1$ and a path $L$-coloring $c_2$ of $G_2$ such that $c_1(u_2) = c_2(u_2)$. Let $c$ be the $L$-coloring of $G$ defined by $v \mapsto c_1(v)$ for $v \in G_1$ and $v \mapsto c_2(v)$ for $v \in G_2$. Observe that $\deg_c(v) = \deg_{c_1}(v)$ for each $v \in G_1 - u_2$, $\deg_c(v) = \deg_{c_2}(v)$ for each $v \in G_2 - u_2$, and $\deg_c(u_2) = \deg_{c_1}(u_2) + \deg_{c_2}(u_2)$. To show that $c$ is a path $L$-coloring of $G$ it suffices to show that $\deg_c(u_2) \leq 2$. We'll also need to show that the $\deg_c$ constraints hold for $x, y, z$, but these will often follow immediately from the inductive hypothesis.

**Case 3.2.1.** Suppose that $u_2 \in C[x, z]$. Observe that $x, y, z \in C_2$. Apply the inductive hypothesis to find a path $L$-coloring $c_2$ of $G_2$ with $x_2 = x$, $y_2 = y$, and $z_2 = z$. Define $L'(u_2) = \{c_2(u_2)\}$ and $L'(v) = L(v)$ for each $v \in G_1 - u_2$. Find a path $L'$-coloring $c_1$ of $G_1$ with $x_1 = y_1 = z_1 = u_2$. Since $\deg_{c_1}(u_2) = 0$, $c$ is a path $L$-coloring of $G$. Moreover, $c_2$, and hence $c$, satisfies the $\deg_c$ constraints of the lemma for $x, y, z$.

**Case 3.2.2.** Suppose that $u_2 \in C[y, x] - x - y$. Pick $d \in L(u_2) - L(z)$. Define $L'(u_2) = \{d\}$ and $L'(v) = L(v)$ for each $v \in G - u_2$. Observe that $x \in C_1$ and $z, y \in C_2$. Apply the inductive hypothesis to find a path $L'$-coloring $c_1$ of $G_1$ with $x_1 = z_1 = x$ and $y_1 = u_2$. Then find a path $L'$-coloring $c_2$ of $G_2$ with $x_2 = u_2$, $y_2 = y$, and $z_2 = z$. Note

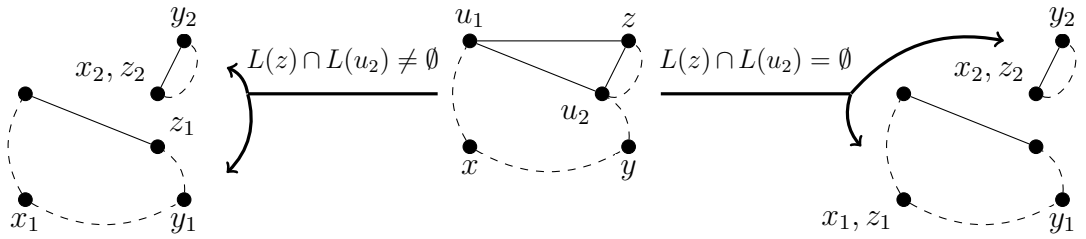



Figure 8: The proof of Lemma 4.1, Case 3.2.3.

that $\deg_{c_1}(u_2) \leq 1$ and $\deg_{c_2}(u_2) \leq 1$, thus $c$ is a path $L$-coloring of $G$. Since $x_1 = x$ in $G_1 - u_2$, and $y_2 = y$ and $z_2 = z$ in $G_2 - u_2$, $c$ satisfies the $\deg_c$ constraints of the lemma.

**Case 3.2.3.** Suppose that $u_2 \in C[z, y]$. Note that $z \neq y$. There are two cases to consider, see Figure 8.

**Case 3.2.3.1.** Suppose that $L(z) \cap L(u_2) \neq \emptyset$. Then either $u_2 = y$ and $L(z) = L(y)$, or $C[z, y] \neq z, y$. Define $L'(u_2) = L(z)$ and $L'(v) = L(v)$ for each $v \in G - u_2$. Find a path $L'$-coloring $c_1$ of $G_1$ with $x_1 = x$, $y_1 = y$, and $z_1 = u_2$. Then find a path $L'$-coloring $c_2$ of $G_2$ with $x_2 = u_2$, $y_2 = z$, and $z_2 = u_2$. Since $\deg_{c_1}(u_2) \leq 1$ and $\deg_{c_2}(u_2) = 1$, it follows that $c$ is a path $L$-coloring of $G$. If $u_2 = y$, then $z_1 = y_1 = y$ and $\deg_{c_1}(y) = 0$; thus $\deg_c(y) = 1$. So $\deg_c(y) \leq 1$ and $c$ satisfies the constraints of the lemma.

**Case 3.2.3.2.** Suppose that $L(z) \cap L(u_2) = \emptyset$. Find a path $L$-coloring $c_1$ of $G_1$ with $x_1 = x$, $y_1 = y$, and $z_1 = x$. Define $L'(u_2) = \{c_1(u_2)\}$ and $L'(v) = L(v)$ for each $v \in G_2 - u_2$. Then find a path $L'$-coloring $c_2$ of $G_2$ with $x_2 = u_2$, $y_2 = z$, and $z_2 = u_2$. Because $C[z_2, y_2] = z_2, y_2$ and $L'(z_2) \cap L'(y_2) = \emptyset$, we must have $\deg_{c_2}(u_2) = \deg_{c_2}(z_2) = 0$. Thus $c$ is a path $L$-coloring of $G$.

Suppose $C[z, y] = z, y$. Then it must be that $u_2 = y$ and $k = 2$. Therefore $G_2 = z, y$ and $\deg_c(z) = 0$, since $c(z) \notin L(y)$. Thus $c$ satisfies the constraints of the lemma. □

Let $G$ be a planar graph and $L$ a list assignment such that $|L(v)| \geq 3$ for each $v \in G$. Compute a planar embedding and add edges to produce a triangulated plane graph $G'$. Pick $u \in G'$ on the outer face and $d \in L(u)$. Define the list assignment $L'$ by $L'(u) = \{d\}$ and $L'(v) = L(v)$ for $v \in G' - u$. By Lemma 4.1 there exists a path $L'$-coloring of $G'$, and thus a path $L$-coloring of $G$. So Theorem 1.2 follows immediately from Lemma 4.1.

The proof of Lemma 4.1 describes a linear time path list-coloring algorithm, see Algorithm 4.2. For $v \in G$ and $u \in N(v)$, we will assume that $\text{next}(^v u)$, $\text{prev}(^v u)$, and $^u v$ can be computed in constant time from $^v u$, see augmented adjacency lists in Section 2.

**Algorithm 4.2:** Path 3-Choose

**Input.** Let $C$ be a cycle or length 1 path in a weakly triangulated plane graph $G$ represented via augmented adjacency lists. Let $x, y, z \in C$ be such that $z \in C[x, y]$.

Let $L$ be a length $n$ array of lists of colors such that for $v \in \{x, y, z\}$ each list $L(v)$ has length one, for $v \in C - \{x, y, z\}$ each list $L(v)$ has length two or three, and for $v \in \text{int}(C)$ each list $L(v)$ has length three. Furthermore, for each $v \in C[x, z] - z$, assume that $L(v)$ does not contain the color in $L(z)$.

Let $A, B$ be length $n$ arrays such that for each $v \in C$ we have $A(v) = {}^{v}a$ and $B(v) = {}^{v}b$, where $a$ is the vertex immediately clockwise from $v$ around $C$, and $b$ is the vertex immediately counter clockwise from $v$ around $C$.

Let $S$ be a length $n$ integer array such that, for $v \in \overline{\text{int}}(C)$, $S(v) \neq 0$ if and only if $v \in C$, and $S(v) = S(x)$ if and only if $v \in C[x, z] - z$ (or $v = x = z$).

Let $\max(S) = \max_{v \in G} S(v)$ and $m_C = m(\overline{\text{int}}(C))$. Define $\ell = \max(S) + r$, where $r$ is determined by the procedure case below that the input satisfies. If $m_C = 1$ (Base case) or $z \neq x$ (Case 3), then $r = 0$. If $z = x$ and $C[y, x] \neq y, x$ (Case 2), then $r = 1$. Otherwise $z = x$ and $C[y, x] = y, x$ (Case 1), and $r = 2$. Assume that $\ell < 2(m - m_C) + 4$.

Let $M$ be a length $2m + 2$ integer array such that if $v \in C$, then $M(S(v)) = S(y)$ if and only if $v \in C[z, y]$.

The concrete input will be $G$, $L$, $A$, $B$, $S$, $M$, and the tuple $(x, y, z)$.

**Output.** For each $v \in \overline{\text{int}}(C)$, all but one color will be removed from the list $L(v)$. The remaining color in each list will represent a path $L$-coloring of $\overline{\text{int}}(C)$ such that $\deg_c(x) \leq 1$, $\deg_c(y) \leq 1$, and $\deg_c(z) \leq 1$. If $z = y$, or if $C[z, y] = z, y$ and $L(z)$ did not contain the color in $L(y)$, then $\deg_c(z) = 0$. Moreover, afterwards $\max(S) \leq \ell + 2(m_C - 1)$.

**Procedure.** Let ${}^{z}u_1 := B(z)$. Our strategy will follow the proof of Lemma 4.1.

**Base case.** If $A(z) = B(z)$, then $C = z, u_1$ is a length 1 path. If $u_1 \neq x$ and $u_1 \neq y$, remove the color in $L(z)$ from $L(u_1)$, if it exists, then remove further colors until one color remains. Note that $m_C = 1$ and $\max(S) \leq \ell + 2(m_C - 1) = \ell$ by definition.

**Recursive case.** Suppose that $A(z) \neq B(z)$. Let ${}^{z}u_2 := \text{next}({}^{z}u_1)$. It follows immediately that $\max(S) \leq \ell + 2(m_C - 1)$ after Cases 1, 2, and 3.1, so we will only argue that this bound holds after Case 3.2. It may be useful to refer to Figures 6, 7, and 8.

**Case 1.** Suppose that $u_1 = y$. Then $z = x$, since $z \in C[x, y]$. Set $S(x) \leftarrow \max(S) + 1$ and $S(y) \leftarrow S(x)$. Assign $M(S(x)) \leftarrow S(x)$. Make a recursive call with the input $(y, x, y)$.

**Case 2.** Suppose that $z = x$ and $u_1 \neq y$. Remove the color in $L(z)$ from $L(u_1)$, if it exists, then remove further colors until one color remains. Assign $S(u_1) \leftarrow \max(S) + 1$ and $M(S(u_1)) \leftarrow S(u_1)$, and make a recursive call with input $(u_1, y, z)$.

**Case 3.** Suppose that $z \neq x$ and $u_1 \neq y$. Remove $zu_1$ by assigning $A(u_1) \leftarrow \text{prev}({}^{u_1}z)$ and $B(z) \leftarrow \text{next}({}^{z}u_1)$; refer to Figures 6, 7, and 8.

**Case 3.1.** Suppose that $S(u_2) = 0$, i.e. $u_2 \in \text{int}(C)$. Assign $S(u_2) \leftarrow S(x)$, and initialize $B(u_2) \leftarrow {}^{u_2}u_1 = \text{next}({}^{u_2}z)$ and $A(u_2) \leftarrow {}^{u_2}z$, see Figure 6 (right). Remove the color in $L(z)$ from $L(u_2)$. Make a recursive call with the same input $(x, y, z)$.

**Case 3.2.** Suppose that $S(u_2) \neq 0$, i.e. $u_2 \in C$. Just as in Case 3.2 of the proof of Lemma 4.1, we will separately consider the two blocks $G_1$ and $G_2$ of $\overline{\text{int}}(C) - zu_1$. Note that $m_C - 1 = m(G_1) + m(G_2)$.

As we've already removed the edge $zu_1$ by adjusting $B(z)$ and $A(u_1)$, it remains to "split" the neighborhood of $u_2$ at the edge ${}^{u_2}z$ such that the recursive call on $G_1$ considers only the neighbors of $u_2$ counter-clockwise from $z$, and the call on $G_2$ considers $z$ and all neighbors clockwise from $z$; refer to Figure 7 and Figure 8.

Let ${}^{u_2}a := A(u_2)$ and ${}^{u_2}b := B(u_2)$. To represent $G_1$, assign $A(u_2) \leftarrow {}^{u_2}a$ and $B(u_2) \leftarrow \text{next}({}^{u_2}z)$. To represent $G_2$, assign $A(u_2) \leftarrow {}^{u_2}z$ and $B(u_2) \leftarrow {}^{u_2}b$

**Case 3.2.1.** Suppose that $S(u_2) = S(x)$, i.e. $u_2 \in C[x, z]$. Make a recursive call on

$G_2$ with input $(x, y, z)$; afterwards $\max(S) \leq \ell + 2m(G_1) - 2$. Assign $S(u_2) \leftarrow \max(S) + 1$ and $M(S(u_2)) \leftarrow S(u_2)$, and make a recursive call on $G_1$ with input $(u_2, u_2, u_2)$. Finally, observe that $\max(S) \leq \ell + 2m(G_1) - 2 + 1 + 2m(G_2) - 1 < \ell + 2(m_C - 1)$.

**Case 3.2.2.** Suppose that $S(u_2) \neq 0$, but $S(u_2) \neq S(x)$ and $M(S(u_2)) \neq S(y)$. In other words, suppose that $u_2 \in C[y, x] - x - y$. First remove the color in $L(z)$ from $L(u_2)$, if it exists, then remove colors until only one remains. Set $S(u_2) \leftarrow S(x)$. Make a recursive call on $G_1$ with input $(x, u_2, x)$; afterwards $\max(S) \leq \ell + 2m(G_1)$. Set $S(u_2) \leftarrow \max(S) + 1$ and $M(S(u_2)) \leftarrow S(u_2)$, and make a recursive call on $G_2$ with input $(u_2, y, z)$. Finally, observe that $\max(S) \leq \ell + 2m(G_1) + 1 + 2m(G_2) - 2 < \ell + 2(m_C - 1)$.

**Case 3.2.3.** Suppose that $M(S(u_2)) = S(y)$, i.e. $u_2 \in C[z, y]$. There are two cases to consider, see Figure 8.

**Case 3.2.3.1.** Suppose that the color in $L(z)$ is in $L(u_2)$. Remove all colors from $L(u_2)$ except for the color in $L(z)$. Make a recursive call on $G_1$ with input $(x, y, u_2)$; afterwards $\max(S) \leq \ell + 2m(G_1) - 2$. Assign $S(z) \leftarrow \max(S) + 1$, $S(u_2) \leftarrow S(z)$, and $M(S(z)) \leftarrow S(z)$. Make a recursive call on $G_2$ with input $(u_2, z, u_2)$. Finally, observe that $\max(S) \leq \ell + 2m(G_1) - 2 + 1 + 2m(G_2) - 1 < \ell + 2(m_C - 1)$.

**Case 3.2.3.2.** Suppose that the color in $L(z)$ is not in $L(u_2)$. Assign $M(S(x)) \leftarrow S(y)$. Make a recursive call on $G_1$ with input $(x, y, x)$; afterwards $\max(S) \leq \ell + 2m(G_1)$. Assign $S(z) \leftarrow \max(S) + 1$, $S(u_2) \leftarrow S(z)$, and $M(S(z)) \leftarrow S(z)$. Make a recursive call on $G_2$ with input $(u_2, z, u_2)$. Observe $\max(S) \leq \ell + 2m(G_1) + 1 + 2m(G_2) - 1 = \ell + 2(m_C - 1)$.

Note that this is the only case where $M(S(v))$ is set a value other than $S(v)$. Setting $M(S(x)) \leftarrow S(y)$ effectively "joins" $C[x, u_1]$ onto $C[u_2, y]$ to represent $C_1[z_1, y_1]$ in $G_1$, without having to re-mark each vertex in $C[x, u_1]$ with $S(y)$; see Figure 8.

Let $G$ be a triangulated plane graph represented by augmented adjacency lists. Let $L$ be a length $n$ array of color lists of length 3, representing a list assignment of $G$. We may path $L$-color $G$ using Algorithm 4.2 as follows.

Create length $n$ arrays $A, B$ and initialize $A(v)$ and $B(v)$ for each $v \in C$ on the outer face of $G$. Create a length $n$ integer array $S$ and initialize $S(v) \leftarrow 1$ for each $v \in C$ and $S(v) \leftarrow 0$ for each $v \in G - C$. Pick any $u \in C$, assign $S(u) \leftarrow 2$, and remove all but one color from $L(u)$. Create a length $2m + 2$ array of integers $M$ and initialize $M(i) \leftarrow i$ for each $i \in \{0, 1, 2\}$. Finally, apply Algorithm 4.2 with the input $G$, $L$, $A$, $B$, $S$, $M$, and $(u, u, u)$. The remaining color in each list in $L$ will represent a path $L$-coloring of $G$.

Each case in Algorithm 4.2 performs a fixed number of constant time operations. Case 3 removes an edge from the subgraph(s) under consideration. Case 1 makes a recursive call with an input that satisfies Case 2, and Case 2 makes a recursive call with an input that satisfies Case 3. Therefore the total number of recursive calls made, and thus the overall time complexity, is $\mathcal{O}(n)$. See Figures 9 and 10 for a concrete example.

# 5 Experimental Performance

A C implementation of each algorithm is available [3]. For each $i$ from $4 \ldots 7$, a uniform random sample of 100 plane triangulations of order $10^i$ was generated using the method

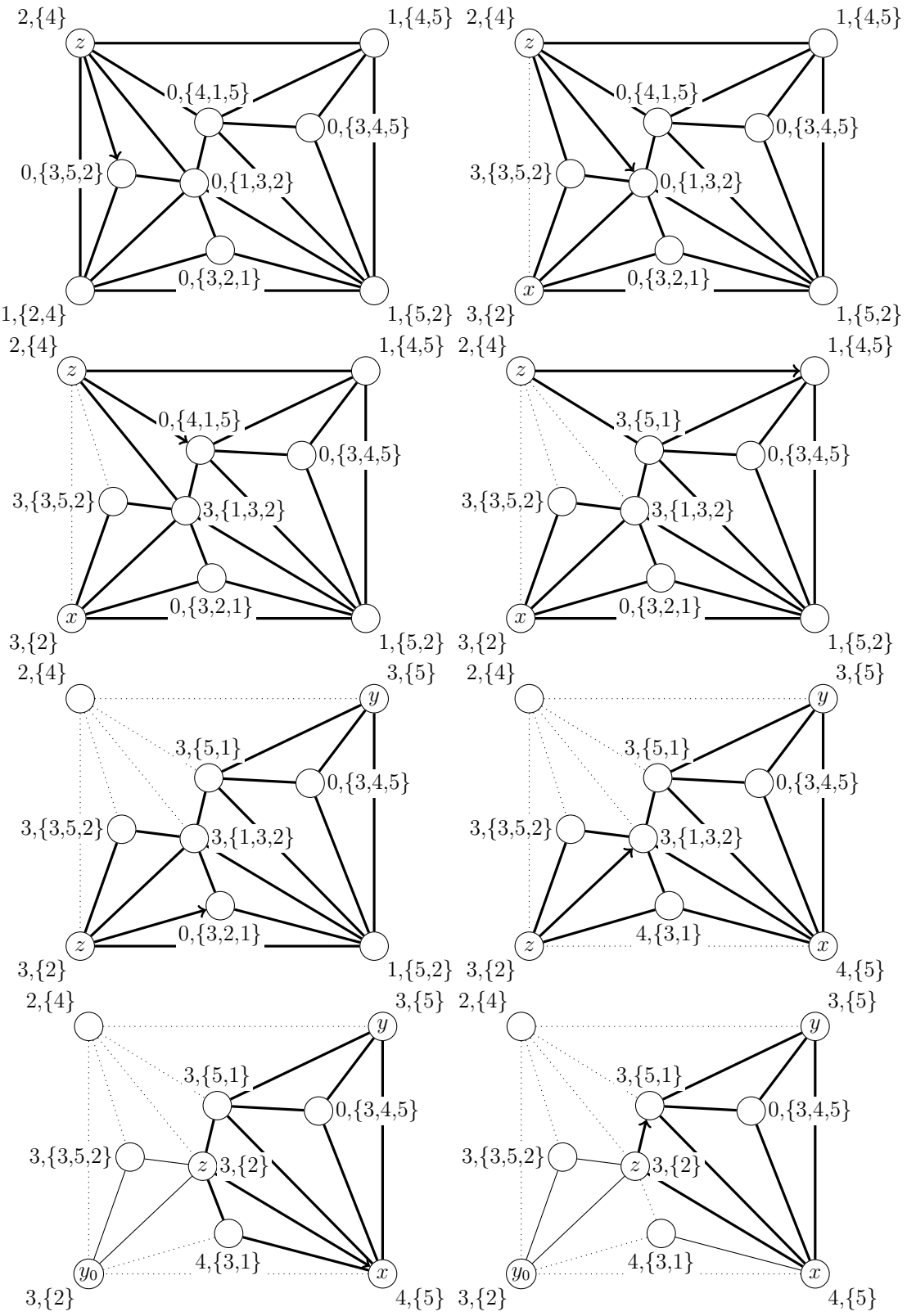



Figure 9: An example of Algorithm 4.2. Each vertex $v \in G$ is labeled with $S(v), L(v)$.

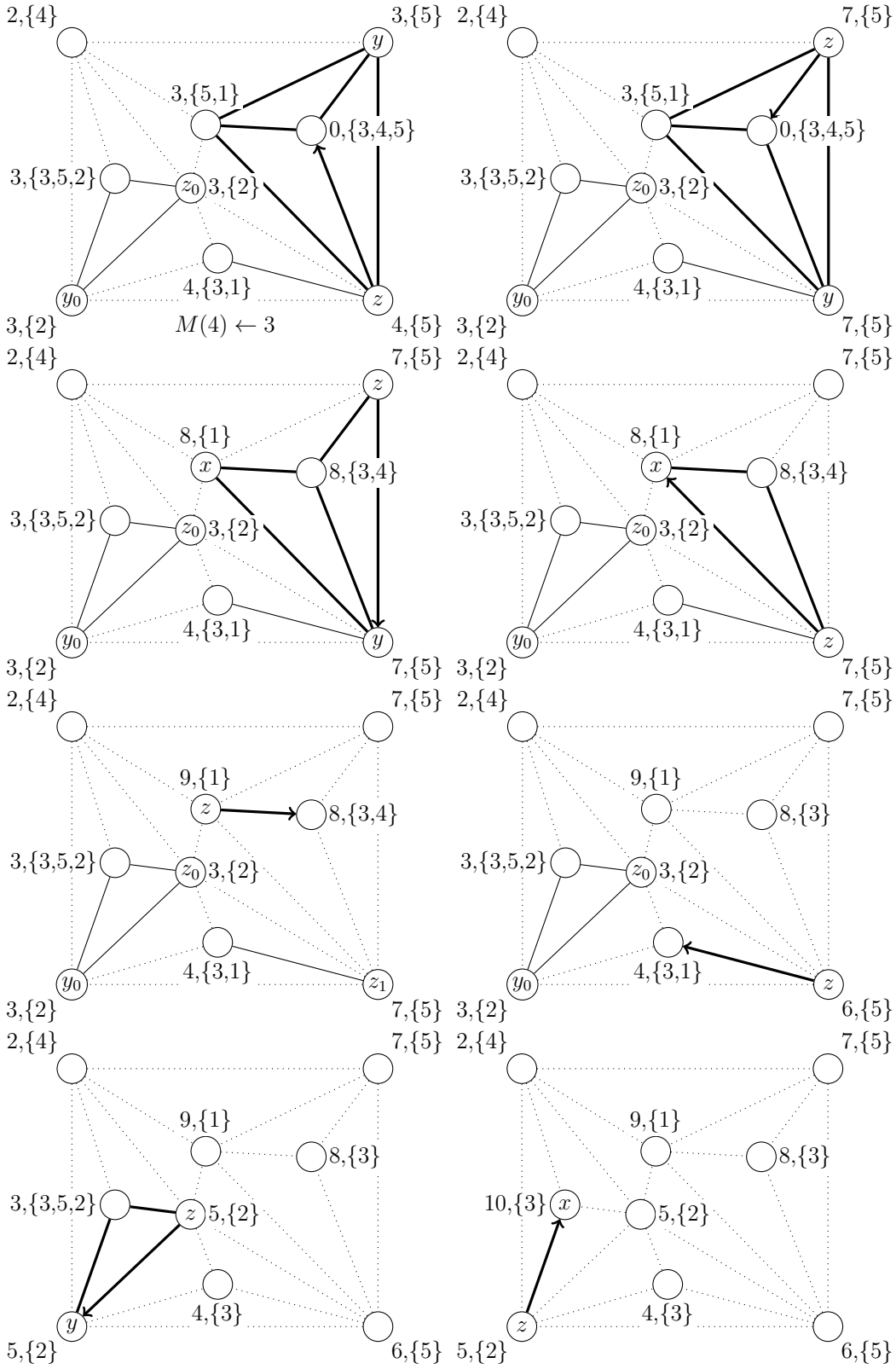



Figure 10: An Algorithm 4.2 example, continued from Figure 9.

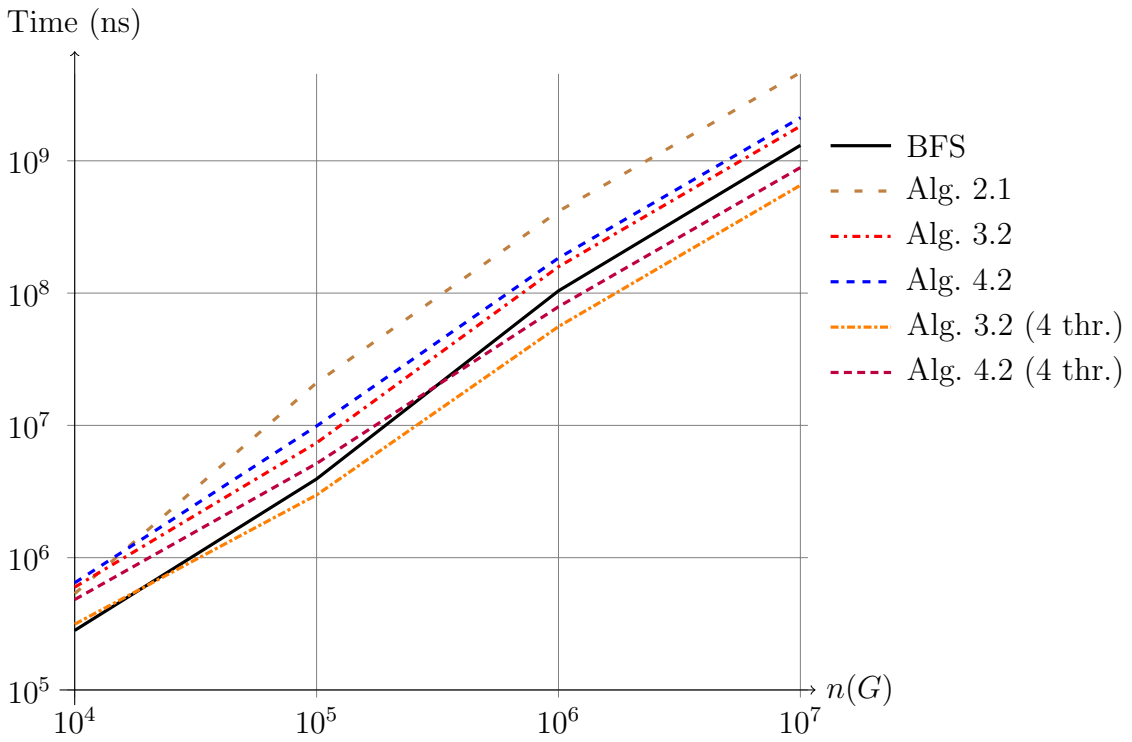



Figure 11: Algorithm benchmarks on a uniform sample of plane triangulations.

of Poulalhon and Schaeffer [10]. Each algorithm, as well as a baseline breadth-first search (BFS), was run on the sample set; see Figure 11.

In Algorithm 3.2 and Algorithm 4.2, it is possible to maintain a stack of independent recursive frames, and to have a pool of threads execute them concurrently. The amount of parallelism possible using this method varies greatly depending on the graph structure, but a 2-3x performance gain was observed when using 4 threads to color large random triangulations; see Figure 11.

Our parallel implementation of Algorithm 3.2 maintains a shared lock-guarded stack which all threads may push/pop recursive frames to/from. To reduce lock contention, each thread also maintains a small local stack of frames, and only access the shared stack when this local stack is full or empty.

Algorithm 4.2 is trickier to adapt to parallel execution: the recursive calls in Case 3.2 rely on a sequential execution order. If we save a local copy of $S(v), A(v), B(v)$ for each $v \in \{x, y, z\}$ alongside each frame, then the recursive calls in Case 3.2.1 with $u_2 = x$, Case 3.2.2, Case 3.2.3.1, and Case 3.2.3.2 with $u_2 = y$ become order independent. However, in Case 3.2.1 with $u_2 \neq x$, and Case 3.2.3.2 with $u_2 \neq y$, the second call will always depend on the color list $L(u_2)$ produced by the first.

Our parallel implementation of Algorithm 4.2 maintains a lock-guarded object pool of frames and a stack of references to frames in the pool. Frames are added to the pool in the usual course of the algorithm, but only pushed to the stack if they are ready to

execute. For each $v \in G$, we store an optional reference to a frame that will become ready when $v$ has been colored. To allow for multiple frames to await the coloring of one vertex, each frame in the pool stores an optional reference to the next frame in line (an intrusive linked list). When a vertex $v$ is colored, each frame waiting on $v$ is pushed to the stack.

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